

this failure and still consider the path valid.

3.5.6.1. Last Known Good UTC Time

Nodes SHALL maintain a stored Last Known Good UTC Time. This time is used as a fallback for cryptographic credentials expiry enforcement, if all other available time synchronization mechanisms fail.

The last known good UTC time SHALL be updated at commissioning and MAY be updated after a successful time synchronization, or by an embedded time in an OTA. Nodes SHOULD store a Last Known Good UTC Time value to persistent storage at least once a month. A Node's initial out-of-box Last Known Good UTC time SHALL be the compile-time of the firmware.

A Node MAY adjust the Last Known Good UTC Time backwards if it believes the current Last Known Good UTC Time is incorrect and it has a good time value from a trusted source. The Node SHOULD NOT adjust the Last Known Good UTC to a time before the later of:

- The build timestamp of its currently running software image
- The not-before timestamp of any of its operational certificates (see [Section 6.4.5, “Node Operational Credentials Certificates”](#)).

If a Node has used the Last Known Good UTC Time, it SHOULD recheck its security materials and existing connections if it later achieves time synchronization.

3.6. Data Confidentiality and Integrity

Symmetric block ciphers are used to provide message security.

All unicast and multicast messages between Nodes requiring protection for confidentiality and integrity with data origin authentication SHALL use Authenticated Encryption with Associated Data (AEAD) as primitive to protect those messages.